ENSTA | INSTITUT POLYTECHNIQUE DE PARIS

Internship Report

LTCI – IMAGES team

Project Title

George Margaritis
IPP M2 student

Under the supervision of

Elsa Angelini

Academic Year 2025–2026

ENSTA

Table of Contents

1 Background **3**
1.1 Karst conduits and the RKE project . . . 3
1.2 Data acquisition . . . 3
1.3 From raw data to a point cloud . . . 4
1.4 The unused horizontal sonar . . . 4

2 Introduction **5**

3 Definitions **5**
3.1 Notation . . . 5
3.2 Frames and poses . . . 6
3.3 Sonar measurements . . . 6
3.4 Gaussian process regression . . . 8
3.5 Scaled beta distribution . . . 8

4 MSIS Scan Merging **8**
4.1 Environment model . . . 9
4.2 Algorithm description . . . 10
4.2.1 Environment prior . . . 10
4.2.2 Gaussian process regression and kernel choices . . . 11
4.2.3 Maximum likelihood estimation of the elevation angle . . . 12
4.2.4 Estimation uncertainty . . . 15
4.3 Pseudocode . . . 18

5 Results **20**

6 Conclusions and Recommendations **20**
6.1 Conclusions . . . 20
6.2 Recommendations . . . 20

List of References **21**

1 Background

This project continues a previous internship on the 3D reconstruction of a flooded karst conduit from sonar data (Margaritis, 2025). This section recalls the scientific context, how the data were collected, how the existing point cloud was produced from them, and which part of the data that work left unused.

1.1 Karst conduits and the RKE project

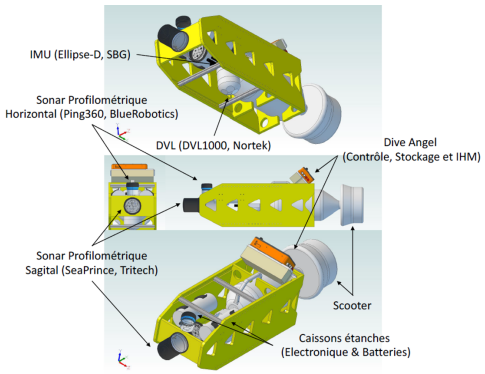
Karst systems form when water dissolves soluble rock such as limestone, creating networks of underground caves and conduits (Ford and Williams, 2007). They store the groundwater used by about a quarter of the world's population (Goldscheider et al., 2020). They are also a hazard: water moves through the conduits quickly and with little filtering, so pollution spreads fast, and heavy rain can overwhelm the network and cause sudden floods. The Fontaine de Nîmes, the site studied here, feeds the flash floods that threaten the city; the flood of 3 October 1988 killed nine people (Maréchal et al., 2008; Batiot-Guilhe et al., 2013). Modelling and managing such an aquifer requires knowing the geometry of its conduits, yet many of them are out of reach for human divers. The *Robots for Karst Exploration* (RKE) project therefore develops robotic platforms to map these flooded networks (Lapierre et al., 2021).

1.2 Data acquisition

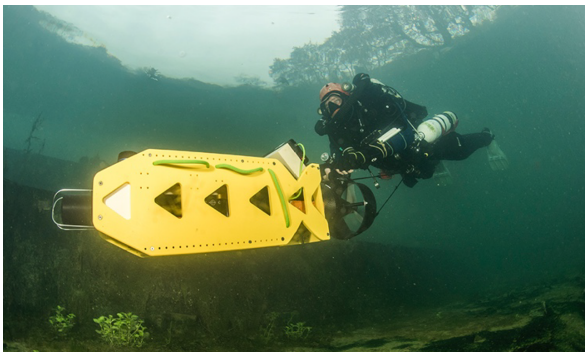
The data were recorded in March 2023 in the flooded conduit of the Fontaine de Nîmes with the NavScoot 2, a diver-propelled vehicle carrying the sensor suite first deployed on the Ulysse robot (Lasbouygues et al., 2017) (Figure 1). Its sensors fall into two groups:

- **Navigation:** a Doppler velocity log (DVL, Nortek DVL 1000) for the velocity, an inertial measurement unit (IMU, SBG Ellipse A) for the orientation, a pressure depth sensor (Blue Robotics Bar30) and a forward-looking echosounder (ISA 500) with its own attitude unit.
- **Mapping:** two mechanically scanned imaging sonars (MSIS). A *vertical* profiling sonar (Tritech Super SeaKing, beam width 1–2°) sweeps a plane across the conduit, measuring its section: roof, floor and walls. A *horizontal* scanning sonar (Blue Robotics Ping360) sweeps a full circle in the horizontal plane, with a beam 2° wide horizontally but 25° high vertically.

An MSIS emits one narrow acoustic ping at a time and rotates its head step by step; each ping returns an intensity profile along a single direction (the *echo*). A full scan is therefore built over several seconds while the vehicle keeps moving.



(a)



(b)

Figure 1: The NavScoot 2 acquisition platform (Margaritis, 2025): (a) the vehicle with its sensors; (b) the diver operating it.

1.3 From raw data to a point cloud

The previous work re-implemented the original MATLAB processing chain in Python and corrected several errors in it (Margaritis, 2025). The pipeline works in three stages:

1. **Trajectory.** DVL velocity spikes above 2 m/s are removed, all sensors are restricted to their common time window and expressed in the vehicle body frame. The velocity is rotated into the world frame with the IMU orientation and integrated over time, while the depth sensor gives the vertical position (dead reckoning).
2. **Echo detection.** In every ping of the vertical sonar, the noise close to the transducer is cut, and the first return above an intensity threshold is kept as the distance to the conduit wall; later returns are discarded as multipath echoes.
3. **Projection.** Each detection (range, beam angle, time) is placed in 3D using the vehicle pose interpolated at the time of the ping, giving a point cloud of about 50 000 points.

Integrating velocities accumulates drift, which distorts the point cloud. It was reduced with the Continuous-time SLAM of the 3D Toolkit (3DTK) (Borrmann et al., 2008; Nüchter et al., 2017), which splits the trajectory into overlapping segments and deforms it smoothly so that the parts of the cloud that see the same walls align (Figure 2). The corrected cloud was then turned into a surface mesh with learning-based reconstruction methods.



Figure 2: Initial point cloud of the conduit obtained from the vertical sonar (Margaritis, 2025).

1.4 The unused horizontal sonar

The point cloud of Margaritis (2025), and everything built on it, uses only the echoes of the vertical sonar. Each vertical scan describes one cross-section of the conduit, so the map is a sequence of slices whose spacing along the conduit depends entirely on the estimated trajectory. The horizontal sonar, recorded during the same dive, sees the walls along the direction of travel and could provide extra information about the conduit geometry, but it was not used in that work. (Figure 3).

Exploiting it is, however, not trivial:

- **Elevation ambiguity:** with a beam 25° high, an echo may come from anywhere within that vertical aperture, so the height of a detected point is uncertain.

- **Motion during a scan:** a 360° sweep takes several seconds, during which the vehicle moves and turns, so each ping must be placed with its own pose.
- **Fusion of perpendicular scans:** the two sonars observe different planes, at different rates and with different noise, and must be expressed consistently in a single frame and point cloud.

The aim of this project is to process the horizontal sonar data and merge them with the vertical sonar data into one consistent point cloud (Section “MSIS Scan Merging”).

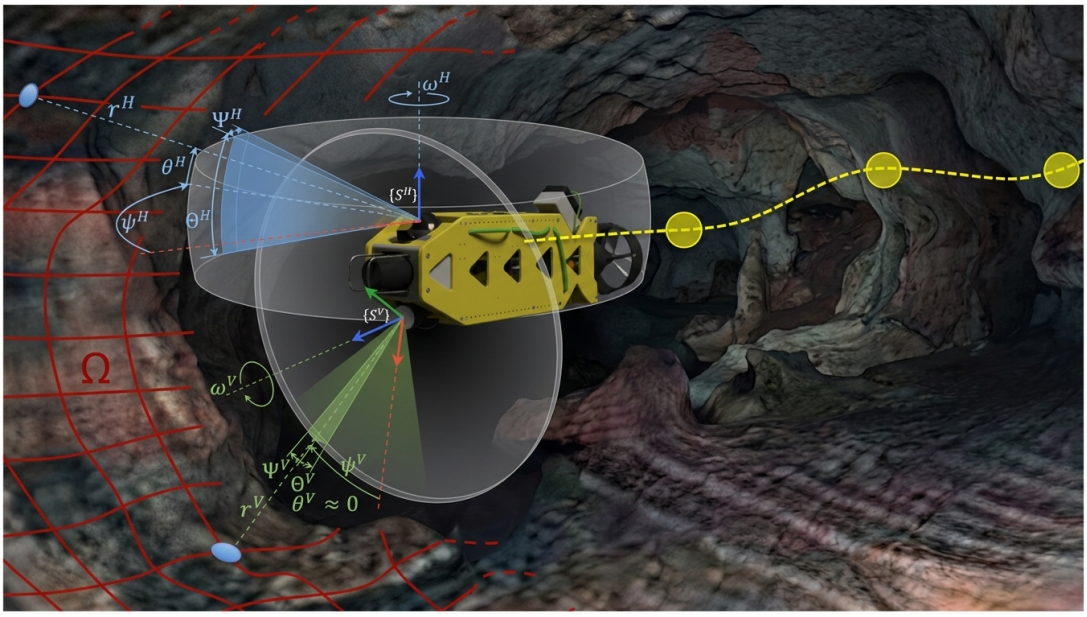


Figure 3: Scanning planes of the two MSIS on the NavScoot 2.

2 Introduction

[Briefly outline the company and/or department you worked for. Summarise the work you have done and clearly state the objectives of your project. You can also state the importance of your project and its relevance.]

3 Definitions

The scan-merging method implemented in this project is the one proposed by Breux and Lapierre (2020) for a vehicle carrying the same pair of perpendicular sonars. This section fixes the notation, frames and models of that work, adapted to the NavScoot 2, and used in the rest of the report.

3.1 Notation

- Vectors are written in bold lower case ($\mathbf{x}$) and matrices in upper case ($\mathbf{M}$). $\mathbf{x}|_{\{R\}}$ denotes a vector expressed in the frame $\{R\}$.
- $X \sim q(\lambda)$ means that the random variable X follows the law of probability density function (pdf) q with parameters λ; $\mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ is the Gaussian law of mean $\boldsymbol{\mu}$ and covariance $\boldsymbol{\Sigma}$. $\hat{X}$ denotes an estimate of X.
- $SE(3)$ is the group of rigid transformations of space. A pose is stored as a position and a unit quaternion $\mathbf{q}$, whose rotation matrix is $\mathbf{R}(\mathbf{q})$, or as the equivalent 4×4 homogeneous matrix $\mathbf{T} = \begin{bmatrix} \mathbf{R} & \boldsymbol{\tau} \\ \mathbf{0} & 1 \end{bmatrix}$.
- $SE(3)$ is a Lie group; its Lie algebra $\mathfrak{se}(3)$ holds the infinitesimal motions, written as 6-vectors $\boldsymbol{\xi} = [\mathbf{v}^T \ \boldsymbol{\omega}^T]^T$ with a translational part $\mathbf{v}$ and a rotational part $\boldsymbol{\omega}$. The logarithm map $\log : SE(3) \to \mathfrak{se}(3)$ returns the

motion that leads from the identity to a given transformation. $T_{\mathbf{A}}SE(3)$ is the tangent space of $SE(3)$ at $\mathbf{A}$ and $\langle\cdot,\cdot\rangle_{\mathbf{A}}$ an inner product on it; e is the identity element.

- Orientations can also be given by Euler angles $[\psi\ \chi\ \phi]^T$ (yaw, pitch, roll), with rotation matrix $\mathbf{R}(\psi,\chi,\phi)$; these angles are written with a δ when they are increments, to distinguish them from the sonar angles ψ^h, ψ^v. $[\boldsymbol{\omega}]_\times$ is the skew-symmetric matrix such that $[\boldsymbol{\omega}]_\times\mathbf{x} = \boldsymbol{\omega}\times\mathbf{x}$.
- $\oplus$ (resp. $\ominus$) is the composition (resp. inverse composition) of poses and points (Blanco, 2010): if $\boldsymbol{\eta}$ is the pose of a frame and $\mathbf{x}$ a point expressed in that frame, $\boldsymbol{\eta}\oplus\mathbf{x}$ is the same point expressed in the frame in which $\boldsymbol{\eta}$ is given. Conversely, $\mathbf{x}\ominus\boldsymbol{\eta}$ expresses in the frame of pose $\boldsymbol{\eta}$ a point $\mathbf{x}$ given in the reference frame.

3.2 Frames and poses

Four frames are used (Figure 4a): the fixed world frame $\{0\}$, the body frame $\{B\}$ attached to the vehicle, and the frames $\{S_H\}$ and $\{S_V\}$ attached to the horizontal and vertical sonar heads. With these frames:

- $\boldsymbol{\tau}$ (3×1) and $\mathbf{q}$ (4×1) are the position and attitude of $\{B\}$ in $\{0\}$, and $\boldsymbol{\eta} = [\boldsymbol{\tau}^T\ \mathbf{q}^T]^T$ (7×1) is the *pose* of the vehicle;
- $\mathbf{p}_H = [\boldsymbol{\tau}_H^T\ \mathbf{q}_H^T]^T$ and $\mathbf{p}_V = [\boldsymbol{\tau}_V^T\ \mathbf{q}_V^T]^T$ are the fixed poses of $\{S_H\}$ and $\{S_V\}$ in $\{B\}$, given by the mounting of the sonars on the NavScoot 2 (Margaritis, 2025).

The navigation sensors (DVL, IMU, depth sensor) provide by dead reckoning a pose estimate at regular times, assumed Gaussian, $\boldsymbol{\eta}\sim\mathcal{N}(\bar{\boldsymbol{\eta}},\boldsymbol{\Sigma}_\eta)$. The *discretized trajectory* is the set of these poses, $\mathcal{H} = \{\boldsymbol{\eta}_i\}$, $i\in\{1,\dots,n_r\}$.

3.3 Sonar measurements

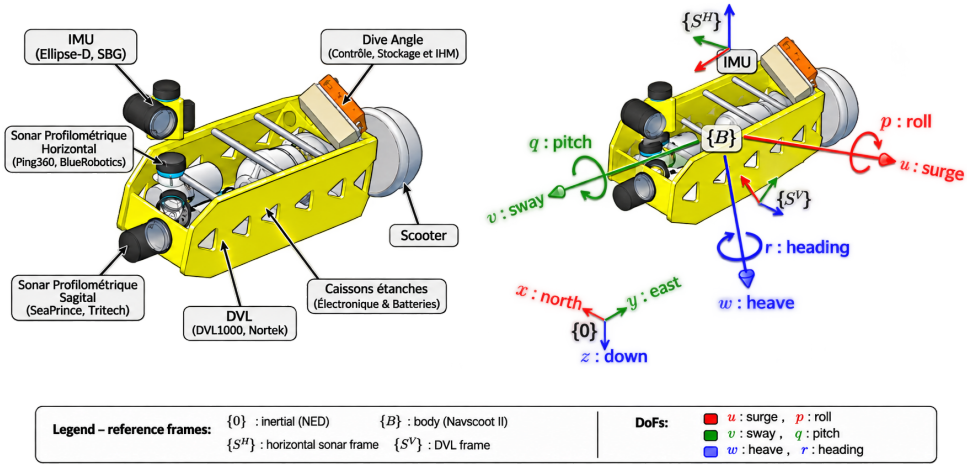
Both sonars are mechanically scanned imaging sonars (MSIS): a head rotates step by step around one axis and, at each step, emits a ping whose echoes give the range ρ to the surrounding surface Ω. In the sonar frame, a point is described by its range ρ, its *rotation angle* ψ (the angle of the head) and its *elevation angle* θ, defined as the oriented angle between the XY-plane of the sonar frame and the measured ray. A beam has a vertical aperture Θ and a horizontal aperture Ψ. On the NavScoot 2, the vertical sonar beam (1–2°) and the horizontal width of the horizontal sonar beam (2°) are narrow, so we take $\Psi_H = \Psi_V = \Theta_V = 0$. Only the vertical aperture of the horizontal sonar is kept: $\Theta_H = [-b/2,\ b/2]$, with $b = 25°$ for the Ping360.

- **Vertical sonar.** A ping gives a single point, $\mathbf{m}_k^v = [\rho_k^v\ \ 0\ \ \psi_k^v]^T$ in $\{S_V\}$: its elevation is known and equal to zero.
- **Horizontal sonar.** Because of the aperture Θ_H, a single ping can return n echoes at different ranges ρ_{ij}^h and *unknown* elevation angles θ_{ij}^h, all with the same rotation angle ψ_i^h (Figure 4b). The i-th measurement is the $3\times n$ matrix $\mathbf{M}_i^h = [\mathbf{m}_{ij}^h]$, with $\mathbf{m}_{ij}^h = [\rho_{ij}^h\ \ \theta_{ij}^h\ \ \psi_i^h]^T$.

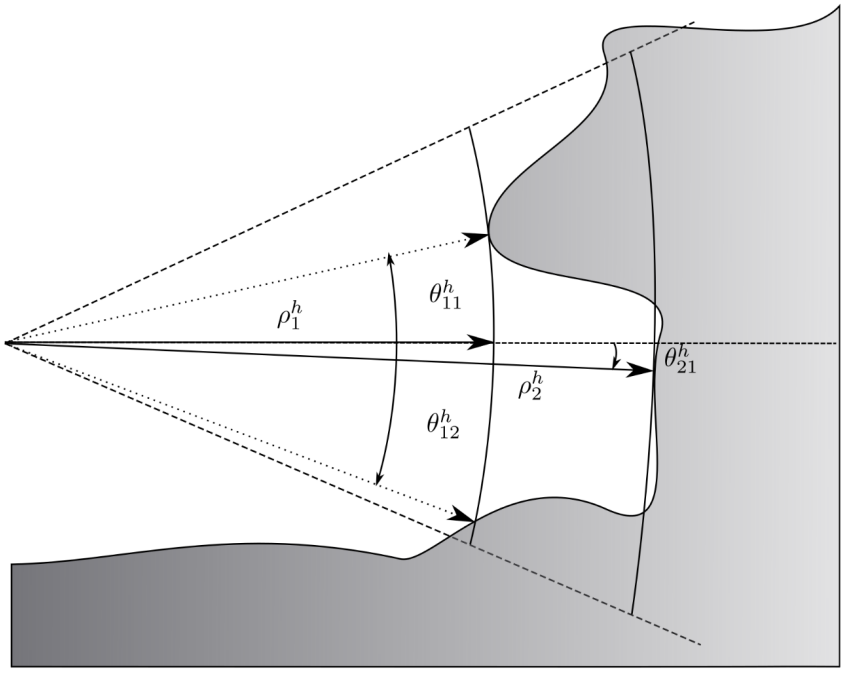
A *complete scan* $\mathcal{S} = \{\mathcal{M}^h,\mathcal{M}^v\}$ gathers the measurements $\mathcal{M}^h = \{\mathbf{M}_i^h\}$, $i\in\{1,\dots,n_h\}$, taken while the horizontal sonar completes a full $360°$ rotation, and the measurements $\mathcal{M}^v = \{\mathbf{m}_k^v\}$, $k\in\{1,\dots,n_v\}$, of the vertical sonar over the same time interval. Since the sonars and the navigation are not synchronized, the functions $\sigma^h:\{1,\dots,n_h\}\to\{1,\dots,n_r\}$ and $\sigma^v:\{1,\dots,n_v\}\to\{1,\dots,n_r\}$ return the index of the pose at which each measurement was taken. A measurement is then placed in the world frame by

$$g^h(\rho,\theta,\psi,i) = \boldsymbol{\eta}_{\sigma^h(i)}\oplus\mathbf{p}_H\oplus\begin{bmatrix}\rho\cos\theta\cos\psi\\ \rho\cos\theta\sin\psi\\ \rho\sin\theta\end{bmatrix}, \tag{1}$$

and g^v is defined in the same way with σ^v and $\mathbf{p}_V$. For a vertical measurement every argument is known. For a horizontal one, θ is missing: estimating it is the goal of scan merging.



(a)



(b)

Figure 4: (a) Frames used in this report. (b) A single horizontal sonar ping can correspond to several 3D points, hence several elevation angles. Adapted from Breux and Lapierre (2020).

3.4 Gaussian process regression

A Gaussian process (GP) is a set of random variables of which every finite subset follows a joint Gaussian law (Rasmussen and Williams, 2006). A GP $f \sim \mathcal{GP}(f_p, K)$ is defined by a prior mean function f_p and a covariance function, or *kernel*, $K(\mathbf{x}, \mathbf{x}')$, which measures how similar the outputs at two inputs are expected to be. GP regression uses it as a non-parametric model of an unknown function. Here it models the conduit surface seen by the vertical sonar: the inputs are the curvilinear abscissa s of the pose along the trajectory (Section "MSIS Scan Merging") and the rotation angle ψ^v, and the output is the range,

$$\rho^v = f(s, \psi^v) + e, \qquad e \sim \mathcal{N}(0, \sigma_n^2). \tag{2}$$

Given N training inputs $\mathbf{X}$ with observed ranges $\mathbf{y}$, the prediction at a new input $\mathbf{x}_*$ is Gaussian, $\rho_* \sim \mathcal{N}(\bar{\rho}_*, \sigma^2_{\rho_*})$, with

$$\bar{\rho}_* = f_p(\mathbf{x}_*) + \mathbf{k}_*^T (\mathbf{K} + \sigma_n^2 \mathbf{I})^{-1} \big(\mathbf{y} - f_p(\mathbf{X})\big), \tag{3}$$

$$\sigma^2_{\rho_*} = K(\mathbf{x}_*, \mathbf{x}_*) - \mathbf{k}_*^T (\mathbf{K} + \sigma_n^2 \mathbf{I})^{-1} \mathbf{k}_*, \tag{4}$$

where $\mathbf{K}$ is the $N \times N$ matrix of kernel values between training inputs and $\mathbf{k}_*$ the vector of kernel values between $\mathbf{x}_*$ and the training inputs. The variance gives the confidence band around the estimated surface.

The kernel sets the smoothness of the estimated surface. The default choice is the radial basis function (RBF) kernel; the Matérn family (Matérn, 1986) generalizes it with a parameter ν that controls smoothness, from the rough exponential kernel ($\nu = 1/2$) to the RBF kernel ($\nu \to \infty$), with $\nu = 3/2$ as a compromise:

$$K_{\mathrm{RBF}}(\mathbf{x}, \mathbf{x}') = \sigma_p^2 \, e^{-\frac{r^2}{2l^2}}, \qquad K_{\exp}(\mathbf{x}, \mathbf{x}') = \sigma_p^2 \, e^{-\frac{r}{l}}, \qquad K^M_{3/2}(\mathbf{x}, \mathbf{x}') = \sigma_p^2 \Big(1 + \tfrac{\sqrt{3}\, r}{l}\Big) e^{-\frac{\sqrt{3}\, r}{l}}, \tag{5}$$

with $r = \|\mathbf{x} - \mathbf{x}'\|$, σ_p^2 the signal variance and l the length scale. The hyperparameters l, σ_p and σ_n are set by hand or by maximum likelihood on the training data. Products and sums of kernels are also kernels, which allows a kernel to be built for a given problem.

3.5 Scaled beta distribution

An estimated elevation angle $\hat{\theta}$ can only lie inside the beam, so its uncertainty is not modelled by a Gaussian, whose support is infinite, but by a beta distribution scaled to $[-b/2,\, b/2]$ (Johnson et al., 1994; Breux and Lapierre, 2020):

$$p(\theta;\, \alpha, \beta, b) = \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha)\,\Gamma(\beta)} \, \frac{(\theta + \frac{b}{2})^{\alpha - 1} \, (\frac{b}{2} - \theta)^{\beta - 1}}{b^{\alpha + \beta - 1}}, \qquad \theta \in \Big[-\frac{b}{2}, \frac{b}{2}\Big], \tag{6}$$

where Γ is the gamma function, which extends the factorial to real and complex numbers,

$$\Gamma(z) = \int_0^\infty x^{z-1} e^{-x}\, dx, \qquad \mathrm{Re}(z) > 0. \tag{7}$$

The shape parameters follow from a mean $\bar{\theta}$ and a variance σ_θ^2:

$$\alpha = (b + 2\bar{\theta}) \, \frac{b^2 - 4(\bar{\theta}^2 + \sigma_\theta^2)}{8b\,\sigma_\theta^2}, \qquad \beta = (b - 2\bar{\theta}) \, \frac{b^2 - 4(\bar{\theta}^2 + \sigma_\theta^2)}{8b\,\sigma_\theta^2}. \tag{8}$$

For $\alpha = \beta = 1$ it reduces to the uniform distribution on the beam, $p = 1/b$, which is the only information available on θ without scan merging.

4 MSIS Scan Merging

The horizontal sonar of the NavScoot 2 measures the range of every echo but not its elevation inside its $25°$ beam (Section "Sonar measurements"). A wide beam is nevertheless valuable: it covers a large part of the conduit at each rotation, which increases the surface seen by two successive $360°$ scans and therefore the chance of recognizing a region already mapped. The price is a high uncertainty on the 3D position of each echo (Breux and

Lapierre, 2020). This project implements the method of Breux and Lapierre (2020) to reduce this uncertainty. Its key idea is that both sonars observe the same environment, which is the variable linking their measurements. The measurements of the high-resolution, narrow-beam vertical sonar are used to build a probabilistic approximation of the surrounding surface by Gaussian process regression; this surface is then used to estimate the probability distribution of the elevation angles of the low-resolution, wide-beam horizontal sonar. In Margaritis (2025), the vertical sonar was only used to add points to the reconstructed environment; here it is also used to estimate the elevation angles of the horizontal sonar.

The method only considers the trajectory during the acquisition of one complete scan (Section "Sonar measurements"): the measurements $\mathcal{M}^h$ of one full rotation of the horizontal sonar, the measurements $\mathcal{M}^v$ of the vertical sonar during the same interval, and the dead-reckoning poses $\mathcal{H} = \{\boldsymbol{\eta}_i\}$, $i \in \{1, \dots, n_r\}$, of that interval. The estimation of the full trajectory, for instance in a pose-graph SLAM framework, is outside the scope of the method.

4.1 Environment model

The surface of the environment is modelled through *virtual vertical sonar measurements*: a 3D point on the surface is parameterized by the 6D pose of the vehicle and the rotation angle of the vertical sonar at which this point could have been seen by the vertical sonar. Formally, the surface is defined as the function

$$f : \begin{cases} SE(3) \times [-\pi, \pi] & \to & \mathbb{R}^+ \\ (\boldsymbol{\eta}, \psi^v) & \mapsto & \rho^v \end{cases} \tag{9}$$

which maps a pose $\boldsymbol{\eta}$ and a rotation angle ψ^v to the distance ρ^v, expressed in the scanning plane of the vertical sonar (the XY-plane of $\{S_V\}$), that would have been measured from the pose $\boldsymbol{\eta}$ with the rotation angle ψ^v. A terrain model of the form $z = f(x, y)$, common in seafloor mapping, cannot be used here: in a conduit, z is not a function of (x, y), since a vertical line crosses the wall both at the floor and at the roof.

The trajectory of the vehicle is a one-dimensional submanifold of $SE(3)$, so it can be parameterized by a single scalar $u \in \mathbb{R}$. Since a kernel measures the similarity of two inputs (Section "Gaussian process regression"), this parameter must be such that, for a kernel K,

$$K(\boldsymbol{\eta}_1, \boldsymbol{\eta}_2) = K(u_1, u_2). \tag{10}$$

The time t does not satisfy Eq. (10): the length scale l of the kernel (Eq. (5)) would depend on time and not on space. For instance, a vehicle standing still at two instants $t_2 \gg t_1$ has $\boldsymbol{\eta}(t_1) = \boldsymbol{\eta}(t_2)$, so that $K(\boldsymbol{\eta}_1, \boldsymbol{\eta}_2) = \sigma_p^2$, whereas $K(t_1, t_2) \approx 0$.

The trajectory is instead parameterized by its *curvilinear abscissa*, i.e. its length in $SE(3)$. Lengths are measured with the left-invariant Riemannian metric used by Breux and Lapierre (2020): for $\mathbf{A} \in SE(3)$ and two tangent vectors $\mathbf{U}_1, \mathbf{U}_2 \in T_{\mathbf{A}}SE(3)$,

$$\langle \mathbf{U}_1, \mathbf{U}_2 \rangle_{\mathbf{A}} = \langle \mathbf{A}^{-1}\mathbf{U}_1, \mathbf{A}^{-1}\mathbf{U}_2 \rangle_e = \mathrm{Tr}(\mathbf{S}_1^T \mathbf{S}_2) = \boldsymbol{\xi}_1^T \mathbf{G}\, \boldsymbol{\xi}_2, \qquad \mathbf{G} = \begin{bmatrix} \mathbf{I}_3 & \mathbf{0} \\ \mathbf{0} & 2\mathbf{I}_3 \end{bmatrix}, \tag{11}$$

where $\mathbf{S}_j = \mathbf{A}^{-1}\mathbf{U}_j \in \mathfrak{se}(3)$ and $\boldsymbol{\xi}_j$ is the same element written as a 6-vector $[\mathbf{v}^T\ \boldsymbol{\omega}^T]^T$. The factor 2 on the rotational part comes from the trace: $\mathrm{Tr}([\boldsymbol{\omega}]_\times^T [\boldsymbol{\omega}]_\times) = 2\|\boldsymbol{\omega}\|^2$. The length of a curve $\boldsymbol{\eta}(t)$ in $SE(3)$ between two instants t_1 and t_2 is then

$$L_{\boldsymbol{\eta}(t_1)}^{\boldsymbol{\eta}(t_2)} = \int_{t_1}^{t_2} \left\langle \frac{d\boldsymbol{\eta}}{dt}, \frac{d\boldsymbol{\eta}}{dt} \right\rangle_{\boldsymbol{\eta}(t)}^{1/2} dt, \tag{12}$$

and the curvilinear abscissa is the length travelled since the start of the scan,

$$s(t) = L_{\boldsymbol{\eta}(0)}^{\boldsymbol{\eta}(t)}. \tag{13}$$

The trajectory available here is the discrete set $\mathcal{H}$ of poses estimated by dead reckoning from the DVL, the IMU and the depth sensor, so the length is approximated by summing the lengths of the relative motions between consecutive poses. With $\mathbf{T}_k$ the 4×4 matrix of the pose $\boldsymbol{\eta}_k$, $s_1 = 0$ and, for $i \in \{2, \dots, n_r\}$,

$$s_i = L_{\boldsymbol{\eta}_1}^{\boldsymbol{\eta}_i} \approx \sum_{k=2}^{i} \langle \Delta\boldsymbol{\eta}_k, \Delta\boldsymbol{\eta}_k \rangle^{1/2} = \sum_{k=2}^{i} \sqrt{\log(\mathbf{T}_{k-1}^{-1}\mathbf{T}_k)^T\, \mathbf{G}\, \log(\mathbf{T}_{k-1}^{-1}\mathbf{T}_k)} = \sum_{k=2}^{i} \left\| \log(\mathbf{T}_{k-1}^{-1}\mathbf{T}_k) \right\|_G, \tag{14}$$

where $\mathbf{T}_{k-1}^{-1}\mathbf{T}_k$ is the motion from $\boldsymbol{\eta}_{k-1}$ to $\boldsymbol{\eta}_k$ (written $\boldsymbol{\eta}_k \ominus \boldsymbol{\eta}_{k-1}$ by Breux and Lapierre 2020), $\log$ the logarithm map of $SE(3)$ and $\|\boldsymbol{\xi}\|_G = \sqrt{\boldsymbol{\xi}^T \mathbf{G}\, \boldsymbol{\xi}}$. The abscissa therefore grows with both the distance travelled and the rotation of the vehicle: a pure rotation of 1 rad adds $\sqrt{2}$ to s, as much as a translation of $\sqrt{2}$ m.
The objective is then to estimate f by training a Gaussian process on the vertical sonar measurements, as a function $f(s, \psi^v)$ of the curvilinear abscissa and the rotation angle. The learned function is finally used to estimate the most likely elevation angles θ_{ij}^h inside the beams of the horizontal sonar.

4.2 Algorithm description

The algorithm is summarized in Algorithm 1 (Section "Pseudocode"). First, the curvilinear abscissa of every pose of the trajectory is computed with Eq. (14) (lines 1–5). The vertical sonar measurements are then placed in the world frame and an elliptic cylinder is fitted to them as the surface prior f_p (lines 6–7, Section "Environment prior"). The Gaussian process modelling the surface is trained on the vertical sonar measurements and their curvilinear abscissae (lines 8–10, Section "Gaussian process regression and kernel choices"). Then, for every echo of every horizontal measurement, the most likely elevation angles are estimated (line 14, Algorithm 2, Section "Maximum likelihood estimation of the elevation angle") and their uncertainties are computed (line 16, Algorithm 3, Section "Estimation uncertainty"). Finally, each estimated angle turns the echo into a 3D point with a known uncertainty, which is added to the vertical sonar points of the scan (line 17). This last step is specific to this project, whose aim is a single point cloud combining both sonars; Breux and Lapierre (2020) stop at the distributions $p(\hat{\theta})$.

4.2.1 Environment prior

The prior f_p is the mean function of the Gaussian process: it encodes what is known of the shape of the environment before the regression, so that the process only has to learn how the real surface deviates from it. Based on the typical geometry of karst conduits, the natural choice is an *elliptic cylinder*. For a pose $\boldsymbol{\eta} = [\boldsymbol{\tau}^T\ \mathbf{q}^T]^T$ on the trajectory (equivalently, its curvilinear abscissa) and a rotation angle ψ^v, $f_p(\boldsymbol{\eta}, \psi^v)$ returns the distance to the prior shape along the corresponding ray of the vertical sonar. Let the cylinder have the pose $\mathbf{T}_c = [\mathbf{R}_c\ \boldsymbol{\tau}_c]$, with its axis along the z-axis of its frame and the semi-axes r_1 and r_2 of its elliptic section along the x- and y-axes. The ray starts at the sonar head $\mathbf{o} = \boldsymbol{\tau} + \mathbf{R}(\mathbf{q})\boldsymbol{\tau}_V$ with the unit direction $\mathbf{u} = \mathbf{R}(\mathbf{q})\mathbf{R}(\mathbf{q}_V)[\cos\psi^v\ \ \sin\psi^v\ \ 0]^T$. Expressed in the frame of the cylinder, $\mathbf{o}' = \mathbf{R}_c^T(\mathbf{o} - \boldsymbol{\tau}_c)$ and $\mathbf{u}' = \mathbf{R}_c^T\mathbf{u}$, and $f_p(\boldsymbol{\eta}, \psi^v)$ is the smallest positive root λ of the polynomial equation of degree two

$$\left(\frac{o'_x + \lambda u'_x}{r_1}\right)^2 + \left(\frac{o'_y + \lambda u'_y}{r_2}\right)^2 = 1. \tag{15}$$

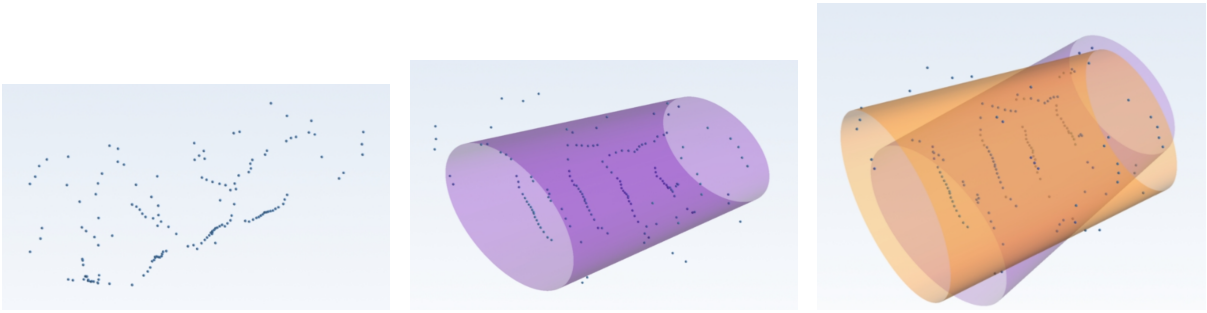
Fitting this primitive to the vertical sonar points $\mathcal{P}^v = \{\mathbf{p}_k = g^v(\rho_k^v, 0, \psi_k^v, k)\}$ is a non-linear optimization, which, like any such optimization, needs a good initial shape. It is done in three steps (Breux and Lapierre, 2020):

1. **Initial axes.** A robust principal component analysis (RPCA) of $\mathcal{P}^v$ gives its centre $\mathbf{c}$ and three principal vectors $\mathbf{a}_1$, $\mathbf{a}_2$, $\mathbf{a}_3$. For a perfect elliptic cylinder, they correspond to the cylinder axis and the major and minor axes of the ellipse. If the cylinder were longer than wide, its axis would be the principal vector with the largest eigenvalue, but this cannot be assumed here, since one scan covers a short portion of conduit. The axis is instead found from tangent vectors: successive points of the vertical sonar lie on the same cross-section of the conduit, so the vector $\mathbf{t}_k = \mathbf{p}_{k+1} - \mathbf{p}_k$ approximates a tangent to the wall that is orthogonal to the axis, and its projection on the axis is minimal:
$$\mathbf{a}_{\text{axis}} = \arg\min_{\mathbf{a}\in\{\mathbf{a}_1,\mathbf{a}_2,\mathbf{a}_3\}} \sum_k \langle \mathbf{a}, \mathbf{t}_k\rangle^2. \tag{16}$$
2. **Elliptic section.** The points are projected on the plane spanned by the two other principal vectors, and an ellipse is fitted to these 2D projections. The PCA basis is then rotated around the cylinder axis to align it with the axes of the ellipse, which gives the initial pose $\mathbf{T}_c$ and semi-axes r_1, r_2.
3. **Refinement.** The asymmetric distribution of the vertical sonar points has a large influence on the RPCA result (Figure 5). The cylinder is therefore refined with the Levenberg–Marquardt (LM) algorithm, which finds

the shape minimizing the orthogonal distances $d_\perp$ of the points to the cylinder:

$$\hat{\mathbf{x}} = \arg\min_{\mathbf{x}\in\mathbb{R}^8} \sum_{k=1}^{n_v} d_\perp(\mathbf{p}_k, \mathbf{x})^2, \tag{17}$$

where $\mathbf{x}$ gathers the 6D pose $\mathbf{T}_c \in SE(3)$ of the cylinder and its two semi-axes. The pose is optimized on the manifold $SE(3)$ (Blanco, 2010): each LM step computes an increment in $\mathfrak{se}(3)$, which is mapped back onto $SE(3)$ by the exponential map.



(a) (b) (c)

Figure 5: Elliptic cylinder prior fitted to the vertical sonar points of one scan: (a) the vertical sonar points, (b) the initial cylinder estimated by RPCA, (c) the final cylinder after Levenberg–Marquardt optimization.

4.2.2 Gaussian process regression and kernel choices

Each vertical measurement $\mathbf{m}_k^v$ provides one training input, the abscissa of its pose and its rotation angle $\mathbf{x}_k = [s_{\sigma^v(k)} \;\; \psi_k^v]^T$, and one output, its range ρ_k^v. They form the training set $\mathbf{X} = [\mathbf{x}_1 \dots \mathbf{x}_{n_v}]$, $\mathbf{y} = [\rho_1^v \dots \rho_{n_v}^v]^T$ of the Gaussian process $f \sim \mathcal{GP}(f_p, K)$ of Eq. (2). The kernel exploits the cylindrical structure of the surface by combining a kernel on the trajectory abscissa and a kernel on the rotation angle:

$$K_{\text{prod}}\big([s\;\psi^v]^T, [s'\;\psi'^v]^T\big) = K_s(s,s')\,K_\psi(\psi^v, \psi'^v). \tag{18}$$

K_s can be any kernel of Section “Gaussian process regression”, or the Matérn 5/2 kernel (Rasmussen and Williams, 2006)

$$K^M_{5/2}(\mathbf{x},\mathbf{x}') = \sigma_p^2\left(1 + \frac{\sqrt{5}\,r}{l} + \frac{5r^2}{3l^2}\right) e^{-\frac{\sqrt{5}\,r}{l}}, \qquad r = \|\mathbf{x}-\mathbf{x}'\|. \tag{19}$$

For K_ψ, a simple Euclidean distance cannot be used, since the angle is periodic ($-\pi$ and π are the same direction), and the distance chosen must keep the kernel positive definite. Two distances on the circle $\mathbb{S}^1$ can be defined, the chordal and the geodesic distances:

$$d_{\text{chord}}(\psi,\psi') = 2\left|\sin\frac{\psi-\psi'}{2}\right|, \tag{20}$$

$$d_{\text{geo}}(\psi,\psi') = \arccos\big(\cos(\psi-\psi')\big). \tag{21}$$

The chordal distance is the Euclidean distance between the points $(\cos\psi, \sin\psi)$ and $(\cos\psi', \sin\psi')$ of the unit circle, so any kernel remains valid with it. With the geodesic distance only some kernels do; for instance, a C^2-Wendland function defines a valid kernel:

$$W_\pi(t) = \left(1 + \tau\frac{t}{\pi}\right)\left(1 - \frac{t}{\pi}\right)_+^\tau, \qquad 0 \le t \le \pi, \quad \tau \ge 4, \tag{22}$$

where $(x)_+ = \max(x, 0)$. Breux and Lapierre (2020) only consider the exponential and Matérn 5/2 kernels with the chordal distance for K_ψ, i.e. $K_\psi(\psi,\psi') = K\big(d_{\text{chord}}(\psi,\psi')\big)$, and compare the exponential, Matérn 3/2, Matérn 5/2 and RBF kernels for K_s. In their simulations, the best K_s depended on the configuration; for K_ψ, the

exponential kernel gave smaller uncertainties than the unknown environment requires and unrealistic learned length scales, while $K_{5/2}^M \circ d_{\text{chord}}$ gave appropriate ones. The pair $K_s = K_{5/2}^M$, $K_\psi = K_{5/2}^M \circ d_{\text{chord}}$ was retained as a reasonable choice. A good kernel trades confidence against accuracy: the variance must be large enough for the confidence interval to contain the real surface, but too large a variance leads to uniform distributions of the elevation angles. Since the surface is estimated per scan, the kernels can also be adapted during the exploration according to prior knowledge of the environment.

The product kernel has one signal variance σ_p^2 and two length scales, l_s along the trajectory and l_ψ on the angle. These hyperparameters and the noise variance σ_n^2 can be set manually or estimated by maximum likelihood on the training set, by maximizing the log marginal likelihood (Rasmussen and Williams, 2006)

$$\ln p(\mathbf{y} \mid \mathbf{X}) = -\frac{1}{2}\big(\mathbf{y} - f_p(\mathbf{X})\big)^T (\mathbf{K} + \sigma_n^2 \mathbf{I})^{-1} \big(\mathbf{y} - f_p(\mathbf{X})\big) - \frac{1}{2} \ln \left|\mathbf{K} + \sigma_n^2 \mathbf{I}\right| - \frac{n_v}{2} \ln 2\pi. \tag{23}$$

The standard regression uses all training inputs and treats them as noise-free. Since the surface is only modelled locally, from the vertical measurements of one scan, the training set is small and a sparse formulation of the Gaussian process is not needed. However, the dead-reckoning drift accumulated during the scan makes the inputs uncertain, and taking this input uncertainty into account could significantly improve the estimation; as in Breux and Lapierre (2020), the inputs are assumed noise-free here.

4.2.3 Maximum likelihood estimation of the elevation angle

The 3D points observed by the horizontal sonar cannot be obtained directly, since any point in its beam may return the same measurement. The surface estimated from the vertical sonar is exploited to infer the elevation angle of each horizontal measurement: elevation angles whose 3D points lie close to the estimated surface are good estimates (Figure 6). For a horizontal measurement (ψ_k^h, ρ_k^h) (one echo of $\mathbf{M}_k^h$), the elevation angle is the solution of

$$\hat{\theta} = \arg\max_{\theta \in \Theta_H} p\big(\hat{\rho}^v \mid f, \psi_k^h, \rho_k^h, \boldsymbol{\eta}_{\sigma^h(k)}, \theta\big) = \arg\max_{\theta \in \Theta_H} \ln p\big(\hat{\rho}^v \mid f, \psi_k^h, \rho_k^h, \boldsymbol{\eta}_{\sigma^h(k)}, \theta\big) = \arg\max_{\theta \in \Theta_H} J(\theta; \psi_k^h, \rho_k^h), \tag{24}$$

where $\hat{\rho}^v$ is the range that would have been obtained if this point had been observed by the vertical sonar. Depending on the environment, several angles can correspond to one measurement (Figure 4b). Instead of the single solution of Eq. (24), the set of local maxima of the log-likelihood is searched:

$$\hat{\Theta} = \left\{ \hat{\theta} \in \Theta_H \;\middle|\; \frac{\partial J}{\partial \theta}(\hat{\theta}) = 0 \text{ and } \frac{\partial^2 J}{\partial \theta^2}(\hat{\theta}) < 0 \right\}. \tag{25}$$

Since a closed-form expression of the likelihood with respect to θ is difficult to obtain, the range $\Theta_H = [-b/2,\, b/2]$, with $b = 25°$ the vertical aperture of the Ping360, is sampled uniformly with n_q values $\theta_q = -b/2 + q\,\Delta_\theta$, $q \in \{1, \ldots, n_q\}$, $\Delta_\theta = b/n_q$. Each sample gives the virtual 3D point

$$\mathbf{x}(\theta_q) = g^h(\rho_k^h, \theta_q, \psi_k^h, k) = \boldsymbol{\eta}_{\sigma^h(k)} \oplus \mathbf{p}_H \oplus \begin{bmatrix} \rho_k^h \cos\theta_q \cos\psi_k^h \\ \rho_k^h \cos\theta_q \sin\psi_k^h \\ \rho_k^h \sin\theta_q \end{bmatrix}. \tag{26}$$

To assess the likelihood that this point lies on the surface, it must be expressed with the inputs and output of f: the curvilinear abscissa $\hat{s}_q$, the rotation angle $\hat{\psi}_q^v$ and the range $\hat{\rho}_q^v$ under which the vertical sonar would have seen it (Figure 7). This takes the five steps below, detailed in Algorithm 2 (Section “Pseudocode”).

1. Selection of the trajectory segments.

Let $\mathbf{n}^v|_{\{S_V\}}$ be the rotation axis of the vertical sonar expressed in its own frame, which is also the normal of its scanning plane; since that plane is the XY-plane of $\{S_V\}$, $\mathbf{n}^v|_{\{S_V\}} = [0\ 0\ 1]^T$. A pose $\hat{\boldsymbol{\eta}}_q$ is searched along the trajectory such that $\mathbf{x}(\theta_q)$ belongs to the vertical sonar plane, which requires interpolating the trajectory. The first step is to determine the segments of the trajectory that may contain $\hat{\boldsymbol{\eta}}_q$; several segments can be found. At the i-th pose, the rotation axis expressed in the world frame is

$$\mathbf{n}^v|_{\{0\},i} = \mathbf{R}(\mathbf{q}_i)\, \mathbf{R}(\mathbf{q}_V)\, \mathbf{n}^v|_{\{S_V\}}, \tag{27}$$

and the plane passes through the sonar head, at $\boldsymbol{\tau}_i^V = \boldsymbol{\tau}_i + \mathbf{R}(\mathbf{q}_i)\boldsymbol{\tau}_V$. Breux and Lapierre (2020) measure the distance to the plane from the vehicle origin $\boldsymbol{\tau}_i$; measuring it from the sonar head matters on the NavScoot 2,

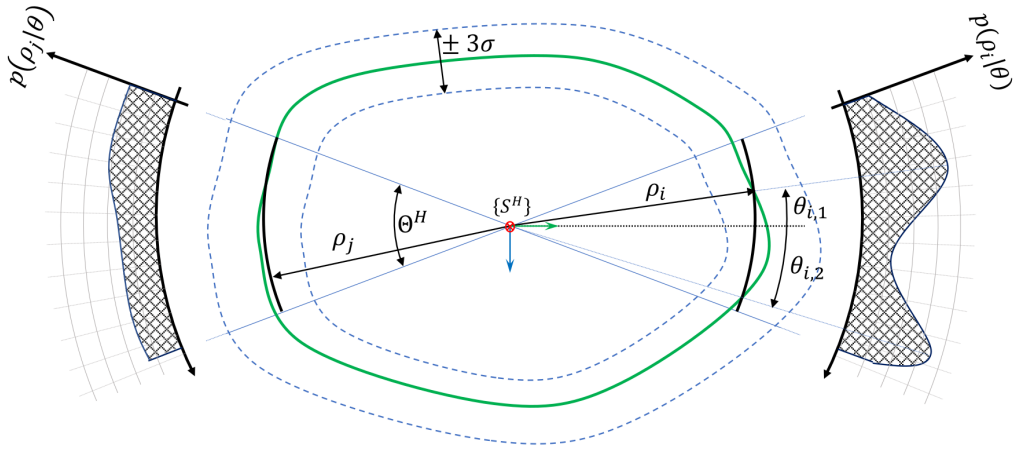


Figure 6: Principle of the elevation angle estimation on a vertical slice: the elevations whose points fall on the estimated surface are the most likely. Adapted from Breux and Lapierre (2020).

whose vertical sonar is mounted 0.45 m ahead of the origin, along the normal of its plane (Margaritis, 2025). A segment $[i, i+1]$ is selected if the projection of $\mathbf{x}(\theta_q) - \boldsymbol{\tau}^V$ on the rotation axis changes sign between the beginning and the end of the segment (Figure 8): by the intermediate value theorem, the segment then contains a pose for which the projection is zero. The set of starting indices is

$$I_q = \left\{ i \;\middle|\; \left\langle \mathbf{n}^v|_{\{0\},i},\, \mathbf{x}(\theta_q) - \boldsymbol{\tau}_i^V \right\rangle \left\langle \mathbf{n}^v|_{\{0\},i+1},\, \mathbf{x}(\theta_q) - \boldsymbol{\tau}_{i+1}^V \right\rangle < 0 \right\}. \tag{28}$$

2. Interpolation of the pose.

For each selected segment, $\boldsymbol{\eta}_{[i,i+1]} : [0,1] \to SE(3)$ denotes the linear interpolation between the poses $\boldsymbol{\eta}_i$ and $\boldsymbol{\eta}_{i+1}$. Its position is

$$\boldsymbol{\tau}_{[i,i+1]}(t) = (\boldsymbol{\tau}_{i+1} - \boldsymbol{\tau}_i)\, t + \boldsymbol{\tau}_i. \tag{29}$$

For the orientation, the relative rotation between the two poses is expressed by its yaw, pitch and roll Euler angles,

$$[\delta\psi \;\; \delta\chi \;\; \delta\phi]^T = \text{Euler}\left(\mathbf{R}(\mathbf{q}_i)^T \mathbf{R}(\mathbf{q}_{i+1})\right), \tag{30}$$

which is the rotational part of $\boldsymbol{\eta}_{i+1} \ominus \boldsymbol{\eta}_i$. The interpolated rotation, the global rotation and the interpolated rotation axis of the vertical sonar are

$$\delta\mathbf{R}(t) = \mathbf{R}(\delta\psi\, t, \delta\chi\, t, \delta\phi\, t), \tag{31}$$

$$\mathbf{R}(t) = \mathbf{R}(\mathbf{q}_i)\, \delta\mathbf{R}(t), \tag{32}$$

$$\mathbf{n}^v|_{\{0\},[i,i+1]}(t) = \mathbf{R}(t)\, \mathbf{R}(\mathbf{q}_V)\, \mathbf{n}^v|_{\{S_V\}}, \tag{33}$$

and the interpolated sonar head is $\boldsymbol{\tau}^V_{[i,i+1]}(t) = \boldsymbol{\tau}_{[i,i+1]}(t) + \mathbf{R}(t)\boldsymbol{\tau}_V$. The interpolation value $\hat{t}$ is the one for which $\mathbf{x}(\theta_q)$ belongs to the sonar plane of normal $\mathbf{n}^v|_{\{0\},[i,i+1]}(\hat{t})$:

$$\left\langle \mathbf{n}^v|_{\{0\},[i,i+1]}(\hat{t}),\; \mathbf{x}(\theta_q) - \boldsymbol{\tau}^V_{[i,i+1]}(\hat{t}) \right\rangle = 0. \tag{34}$$

Assuming small angle variations between consecutive poses, $\delta\mathbf{R}$ is approximated to the first order:

$$\delta\mathbf{R}(t) \approx \begin{bmatrix} 1 & -\delta\psi\, t & \delta\chi\, t \\ \delta\psi\, t & 1 & -\delta\phi\, t \\ -\delta\chi\, t & \delta\phi\, t & 1 \end{bmatrix} = \mathbf{I}_3 + t\,[\delta\boldsymbol{\omega}]_\times, \qquad \delta\boldsymbol{\omega} = [\delta\phi \;\; \delta\chi \;\; \delta\psi]^T. \tag{35}$$

Written in the frame of the vertical sonar, i.e. as the coordinate of $\mathbf{x}(\theta_q)$ along $\mathbf{n}^v|_{\{S_V\}}$ in $\{S_V\}$, Eq. (34) then becomes a polynomial equation of degree two in $\hat{t}$:

$$\mathbf{n}^v|^T_{\{S_V\}}\, \mathbf{R}(\mathbf{q}_V)^T \Big(\delta\mathbf{R}(\hat{t})^T\, \mathbf{R}(\mathbf{q}_i)^T \left((\boldsymbol{\tau}_i - \boldsymbol{\tau}_{i+1})\, \hat{t} + \mathbf{x}(\theta_q) - \boldsymbol{\tau}_i \right) - \boldsymbol{\tau}_V \Big) = 0. \tag{36}$$

With $\mathbf{m} = \mathbf{R}(\mathbf{q}_V)\mathbf{n}^v|_{\{S_V\}}$ the axis in the body frame, $\mathbf{a} = \mathbf{R}(\mathbf{q}_i)^T(\mathbf{x}(\theta_q) - \boldsymbol{\tau}_i)$ and $\mathbf{d} = \mathbf{R}(\mathbf{q}_i)^T(\boldsymbol{\tau}_{i+1} - \boldsymbol{\tau}_i)$, its coefficients are

$$c_2\hat{t}^2 + c_1\hat{t} + c_0 = 0, \qquad c_2 = \mathbf{m}^T[\delta\boldsymbol{\omega}]_\times\mathbf{d}, \quad c_1 = -\mathbf{m}^T\big(\mathbf{d} + [\delta\boldsymbol{\omega}]_\times\mathbf{a}\big), \quad c_0 = \mathbf{m}^T(\mathbf{a} - \boldsymbol{\tau}_V). \tag{37}$$

c_0 is the signed distance to the plane at $\boldsymbol{\eta}_i$ and c_2 is small, being the product of the rotation increment and the displacement, so the equation has a single root in $[0, 1]$ (it is linear if $c_2 = 0$). The interpolated pose $\hat{\boldsymbol{\eta}}_q = \boldsymbol{\eta}_{[i,i+1]}(\hat{t})$ uses the exact rotation of Eqs. (31)–(32); the first-order approximation only serves to find $\hat{t}$.

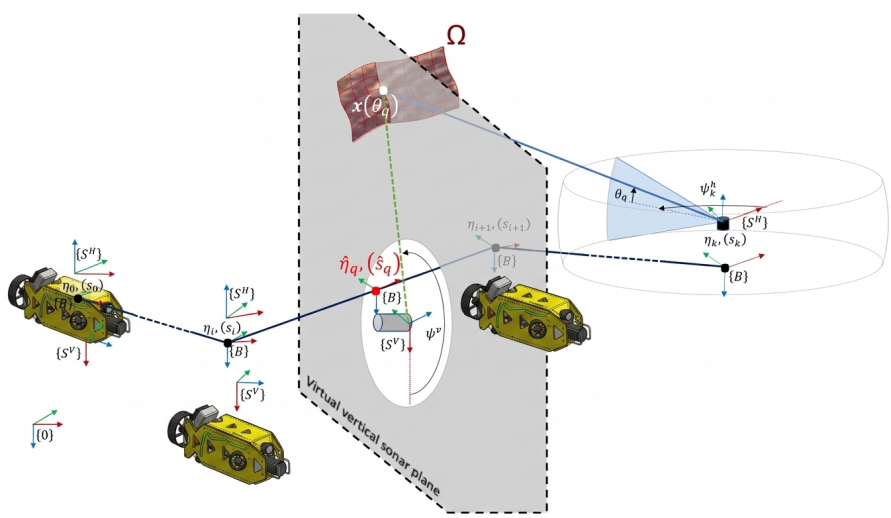


Figure 7: Virtual vertical measurement of a candidate point seen by the horizontal sonar. Adapted from Breux and Lapierre (2020).

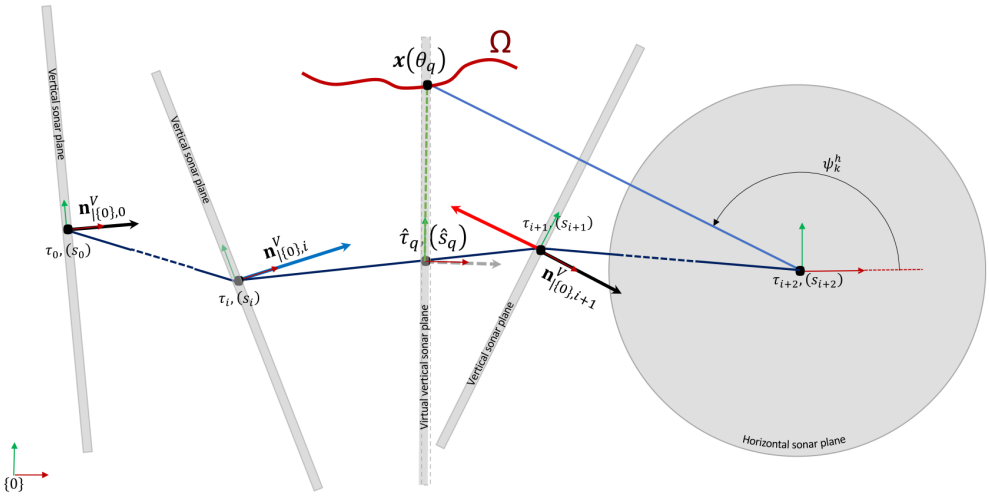


Figure 8: Selection of the trajectory segment containing the pose of the virtual measurement. Adapted from Breux and Lapierre (2020).

3. Virtual vertical measurement.

The curvilinear abscissa corresponding to $\hat{t}$ is

$$\hat{s}_q = s_i + L_{\boldsymbol{\eta}_i}^{\hat{\boldsymbol{\eta}}_q} = s_i + \left\| \log(\mathbf{T}_i^{-1}\hat{\mathbf{T}}_q) \right\|_G, \tag{38}$$

with $\hat{\mathbf{T}}_q$ the matrix of $\hat{\boldsymbol{\eta}}_q$. To compute $\hat{\psi}_q^v$, $\mathbf{x}(\theta_q)$ is expressed in the frame of the vertical sonar at the pose $\hat{\boldsymbol{\eta}}_q$, where its third coordinate is zero by construction of $\hat{t}$:

$$\hat{\mathbf{x}}(\theta_q)|_{\{S_V\}} = \mathbf{x}(\theta_q) \ominus (\hat{\boldsymbol{\eta}}_q \oplus \mathbf{p}_V) = [\hat{x}^v \;\; \hat{y}^v \;\; 0]^T, \tag{39}$$

$$\hat{\psi}_q^v = \operatorname{atan2}(\hat{y}^v, \hat{x}^v), \tag{40}$$

$$\hat{\rho}_q^v = \left\|\hat{\mathbf{x}}(\theta_q)|_{\{S_V\}}\right\|_2. \tag{41}$$

4. Log-likelihood.

The Gaussian process trained to estimate f predicts, at the input $(\hat{s}_q, \hat{\psi}_q^v)$ and with the prior $f_p(\hat{\boldsymbol{\eta}}_q, \hat{\psi}_q^v)$, the range (Eqs. (3)–(4), Rasmussen and Williams 2006, Eqs. 2.25–2.26)

$$\rho_q^* \sim \mathcal{N}\big(\bar{\rho}_q^*, \sigma_q^{*2}\big), \tag{42}$$

and the log-likelihood of the virtual range is

$$J(\theta_q; \psi_k^h, \rho_k^h) = \ln p\big(\hat{\rho}_q^v \mid f, \psi_k^h, \rho_k^h, \boldsymbol{\eta}_{\sigma^h(k)}, \theta_q\big) = -\left(\frac{(\bar{\rho}_q^* - \hat{\rho}_q^v)^2}{2\sigma_q^{*2}} + \ln \sigma_q^* + \frac{\ln(2\pi)}{2}\right). \tag{43}$$

If the trajectory crosses the plane in several segments, the most likely crossing is kept. If no segment is found, $\mathbf{x}(\theta_q)$ lies outside the part of the conduit swept by the vertical sonar during the scan: the sample gets $J = -\infty$ and cannot be a maximum.

5. Local maxima.

Eq. (25) is evaluated on the samples: θ_{q-1} is kept when its log-likelihood is larger than those of its two neighbours θ_{q-2} and θ_q, which approximates $\partial J/\partial\theta = 0$ and $\partial^2 J/\partial\theta^2 < 0$. The resolution of the estimates is therefore $\Delta_\theta = b/n_q$.

4.2.4 Estimation uncertainty

Each estimated angle $\hat{\theta} \in \hat{\Theta}$ also needs an uncertainty, i.e. a probability distribution $p(\hat{\theta})$. A normal distribution is not adapted, as its support is infinite while the angle is restricted to the aperture of the beam. A good choice is the scaled beta distribution (Johnson et al., 1994) of Eq. (6), defined by two parameters α, β,

$$\hat{\theta} \sim \operatorname{Beta}(\alpha, \beta, \theta_{\min}, \theta_{\max}), \qquad \theta_{\max} = -\theta_{\min} = \frac{b}{2}, \tag{44}$$

whose density involves the gamma function of Eq. (7) and whose parameters follow from a mean $\bar{\theta}$ and a variance $\sigma_{\bar{\theta}}^2$ through Eq. (8). The mean is simply the estimate, $\bar{\theta} = \hat{\theta}$. The variance is estimated as the inverse of the Fisher information $\mathcal{I}(\bar{\theta})$:

$$\sigma_{\bar{\theta}}^2 = \frac{1}{\mathcal{I}(\bar{\theta})}, \qquad \mathcal{I}(\theta) = -\mathbb{E}\left[\frac{\partial^2 \ln p(\rho^v \mid \theta)}{\partial\theta^2}\,\middle|\,\theta\right] = -\int \frac{\partial^2 \ln p(\rho^v \mid \theta)}{\partial\theta^2}\, p(\rho^v \mid \theta)\, d\rho^v. \tag{45}$$

The likelihood is the Gaussian prediction of the process at the virtual input of θ, $p(\rho^v \mid \theta) = \mathcal{N}\big(\bar{\rho}(\theta), \sigma_\rho^2(\theta)\big)$, with $\bar{\rho} = \bar{\rho}^*$ and $\sigma_\rho = \sigma^*$ of Eq. (42). Direct computation of Eq. (45) then gives

$$\sigma_{\bar{\theta}}^2 = \frac{\sigma_\rho^2(\bar{\theta})}{2\left(\left.\frac{\partial\sigma_\rho}{\partial\theta}\right|_{\bar{\theta}}\right)^2 + \left(\left.\frac{\partial\bar{\rho}}{\partial\theta}\right|_{\bar{\theta}}\right)^2}. \tag{46}$$

An angle is therefore well determined when a small change of θ moves the predicted range or its uncertainty by much more than that uncertainty.

Derivation of the Fisher information.

Following Appendix A of Breux and Lapierre (2020), write $\bar{\rho}$, σ for $\bar{\rho}(\theta)$, $\sigma_\rho(\theta)$, primes for derivatives with respect to θ, and $u = \rho^v - \bar{\rho}$. The negative log-likelihood and its first two derivatives are

$$-\ln p(\rho^v \mid \theta) = \ln\sigma + \frac{\ln 2\pi}{2} + \frac{u^2}{2\sigma^2}, \tag{47}$$

$$-\frac{\partial \ln p(\rho^v \mid \theta)}{\partial\theta} = \frac{\sigma'}{\sigma} - \frac{u\,\bar{\rho}'}{\sigma^2} - \frac{u^2\sigma'}{\sigma^3}, \tag{48}$$

$$-\frac{\partial^2 \ln p(\rho^v \mid \theta)}{\partial\theta^2} = \frac{\sigma\sigma'' - \sigma'^2}{\sigma^2} + \frac{\bar{\rho}'^2}{\sigma^2} - \frac{u\,\bar{\rho}''}{\sigma^2} + \frac{4u\,\bar{\rho}'\sigma'}{\sigma^3} - \frac{u^2(\sigma\sigma'' - 3\sigma'^2)}{\sigma^4}. \tag{49}$$

Taking the expectation with $\mathbb{E}[u] = 0$ and $\mathbb{E}[u^2] = \sigma^2$, the terms in σ'' cancel and

$$\mathcal{I}(\theta) = \frac{2\sigma'^2 + \bar{\rho}'^2}{\sigma^2}, \tag{50}$$

which is Eq. (46). In practice the derivatives are evaluated by central finite differences on the samples θ_q of Algorithm 2 (one-sided differences when a neighbouring sample has no crossing). When both sonars are at the same place, as in Fig. 6 of Breux and Lapierre (2020), the virtual range $\hat{\rho}^v$ does not depend on θ. On the NavScoot 2 the two sonars are about 0.3 m apart, so $\hat{\rho}^v$ varies slightly with θ; using the derivative of the residual $\bar{\rho}^* - \hat{\rho}^v$ in place of $\bar{\rho}'$ accounts for this.

Validity thresholds.

A uniform distribution is still a beta distribution, with $\alpha = \beta = 1$: from Eqs. (6) and (7),

$$p(\theta; 1, 1, b) = \frac{\Gamma(2)}{b\,\Gamma(1)\,\Gamma(1)} = \frac{1}{b}, \tag{51}$$

the density of the uniform distribution on $[-b/2,\, b/2]$. A threshold σ_l is defined to filter out the variances σ_θ that lead to invalid beta parameters. First, α and β must be strictly positive:

$$\alpha > 0,\ \beta > 0 \iff \sigma_\theta^2 < \left(\frac{b}{2}\right)^2 - \bar{\theta}^2. \tag{52}$$

Furthermore, each distribution is assumed unimodal, so $\alpha \geq 1$ or $\beta \geq 1$: if $\alpha, \beta < 1$, the distribution is U-shaped, with its modes at the edges $\pm b/2$ of the beam. From Eq. (8),

$$\alpha \geq 1 \iff \sigma_\theta^2 \leq A(\bar{\theta}) = \frac{(b + 2\bar{\theta})^2(b - 2\bar{\theta})}{4(3b + 2\bar{\theta})}, \tag{53}$$

$$\beta \geq 1 \iff \sigma_\theta^2 \leq B(\bar{\theta}) = A(-\bar{\theta}), \tag{54}$$

which implies

$$\sigma_\theta^2 \leq \sigma_l^2(\bar{\theta}) = \max\big(A(\bar{\theta}), A(-\bar{\theta})\big) = A(|\bar{\theta}|). \tag{55}$$

The first constraint is included in the second, since $A(|\bar{\theta}|) = \big((b/2)^2 - \bar{\theta}^2\big)\,(b + 2|\bar{\theta}|)/(3b + 2|\bar{\theta}|)$, so that

$$\forall\bar{\theta} \in \left[-\frac{b}{2}, \frac{b}{2}\right], \quad \sigma_l^2(\bar{\theta}) = A(|\bar{\theta}|) \leq \left(\frac{b}{2}\right)^2 - \bar{\theta}^2. \tag{56}$$

A second threshold σ_m filters out estimations that are valid but too uncertain:

$$k\,\sigma_m = \frac{b}{2}, \tag{57}$$

where the parameter k controls the sensitivity of the filter, higher values being more restrictive; Breux and Lapierre (2020) use $k = 3$ by default. The final threshold is

$$\sigma_T^2(\bar{\theta}) = \min\big(\sigma_l^2(\bar{\theta}),\ \sigma_m^2\big). \tag{58}$$

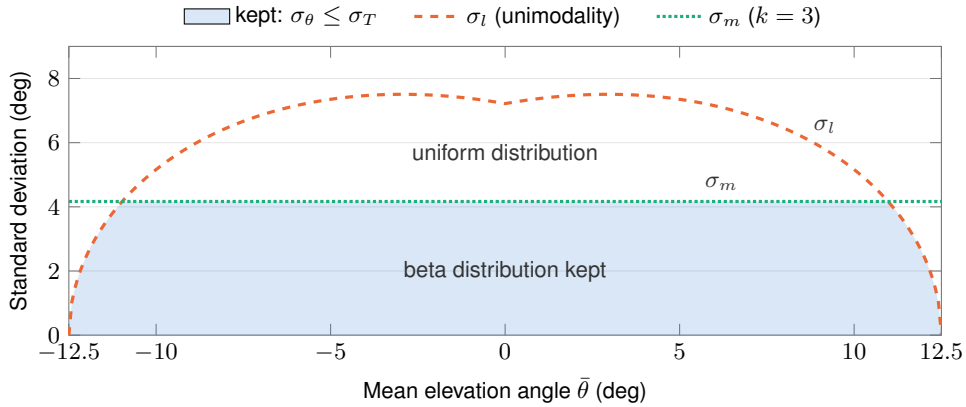


Figure 9: Thresholds on the standard deviation σ_θ of an elevation estimate as a function of its mean $\bar{\theta}$, for the Ping360 ($b = 25°$) and $k = 3$, computed from Eqs. (55)–(58). Estimates below $\sigma_T = \min(\sigma_l, \sigma_m)$ (shaded) keep their beta distribution; the others are replaced by the uniform distribution. Fig. 9 of Breux and Lapierre (2020) shows the same thresholds, as variances, for a $35°$ beam.

If $\sigma_\theta \leq \sigma_T$, α and β are computed with Eq. (8); otherwise the estimate carries no reliable information and its distribution is the uniform one. Figure 9 shows these thresholds for the Ping360 ($b = 25°$) and $k = 3$: $\sigma_m \approx 4.2°$, while σ_l equals the standard deviation of the uniform distribution, $b/\sqrt{12} \approx 7.2°$, at the centre of the beam, peaks at $7.5°$ for $|\bar{\theta}| \approx 3°$ and drops to zero at the edges. Hence $\sigma_T = \sigma_m$ for $|\bar{\theta}| \leq 11°$, and $\sigma_T = \sigma_l$ in the last $1.5°$ of each side of the beam.

To display the result, Breux and Lapierre (2020) colour the measured arc of each echo with the density of its beta distributions, through the mapping

$$m : \begin{cases} \mathbb{R}^+ & \to & [0,1] \\ x & \mapsto & \dfrac{2}{\pi}\arctan(bx) \end{cases} \tag{59}$$

whose central colour, $m = 1/2$, is obtained for $x = 1/b$, the density of the uniform distribution (Figure 10). Algorithm 3 (Section “Pseudocode”) summarizes the computation of the distribution of one estimate.

[One horizontal echo with two local maxima: (a) the measured arc coloured with the mapping of Eq. (59); (b) log-likelihood (top) and inverse Fisher information (bottom) along θ; (c) the two resulting beta distributions (own data, or adapted from Fig. 10 of Breux and Lapierre 2020)]

Figure 10: Uncertainty estimation for a horizontal echo hitting the wall at two elevations.

[Add the settings used on the NavScoot 2 data: how the echoes are extracted from the Ping360 intensity profiles, how the recording is cut into complete scans, the kernels and hyperparameters chosen, the number of samples n_q and the threshold k.]

4.3 Pseudocode

Algorithms 1 to 3 give the complete pseudocode of the method: the scan merging (Algorithm 1), the maximum likelihood estimation of the elevation angles (Algorithm 2, Section “Maximum likelihood estimation of the elevation angle”) and the estimation of their uncertainty (Algorithm 3, Section “Estimation uncertainty”). The line numbers quoted in the text refer to these algorithms.

Algorithm 1 Scan merging (Breux and Lapierre, 2020).

Input: Horizontal measurements $\mathcal{M}^h = \{\mathbf{M}_k^h\}_{k=1}^{n_h}$ and vertical measurements $\mathcal{M}^v = \{\mathbf{m}_k^v\}_{k=1}^{n_v}$ of one complete scan; trajectory $\mathcal{H} = \{\boldsymbol{\eta}_i\}_{i=1}^{n_r}$; sonar poses $\mathbf{p}_H$, $\mathbf{p}_V$; beam aperture b

Output: Distributions $p(\hat{\theta})$ of the elevation angles of every horizontal echo; merged point cloud $\mathcal{P}$

▷ *Curvilinear abscissa*

$s_1 \leftarrow 0$

for $i = 2, \dots, n_r$ **do**

  $\boldsymbol{\xi}_i \leftarrow \log(\mathbf{T}_{i-1}^{-1}\mathbf{T}_i)$ ▷ Relative motion

  $s_i \leftarrow s_{i-1} + \sqrt{\boldsymbol{\xi}_i^T \mathbf{G}\, \boldsymbol{\xi}_i}$ ▷ Eq. (14)

end for

▷ *Environment prior*

$\mathcal{P}^v \leftarrow \{g^v(\rho_k^v, 0, \psi_k^v, k) \mid \mathbf{m}_k^v \in \mathcal{M}^v\}$ ▷ Vertical sonar points in $\{0\}$

$f_p \leftarrow$ FitEllipticCylinder$(\mathcal{P}^v)$ ▷ Eqs. (15)–(17)

▷ *Surface estimation by Gaussian process*

$\mathbf{X} \leftarrow [\,(s_{\sigma^v(k)}, \psi_k^v)\,]_{k=1}^{n_v}$, $\mathbf{y} \leftarrow [\,\rho_k^v\,]_{k=1}^{n_v}$ ▷ Training set

$(\sigma_p, l_s, l_\psi, \sigma_n) \leftarrow \arg\max \ln p(\mathbf{y} \mid \mathbf{X})$ ▷ Eq. (23)

$f \leftarrow \mathcal{GP}(f_p, K_{\text{prod}})$ conditioned on $(\mathbf{X}, \mathbf{y})$ ▷ Eqs. (3), (4), (18)

▷ *Elevation angle estimation and merging*

$\mathcal{P} \leftarrow \mathcal{P}^v$

for $k = 1, \dots, n_h$ **do**

  for each echo (ρ_{kj}^h, ψ_k^h) of $\mathbf{M}_k^h$ **do**

    $\hat{\Theta}, \mathcal{Q} \leftarrow$ ElevationAngleMLE$(f, \mathbf{s}, \rho_{kj}^h, \psi_k^h, k)$ ▷ Algorithm 2

    for $\hat{\theta} \in \hat{\Theta}$ **do**

      $p(\hat{\theta}) \leftarrow$ UncertaintyEstimate$(\mathcal{Q}, \hat{\theta}, b)$ ▷ Algorithm 3

      $\mathcal{P} \leftarrow \mathcal{P} \cup \{(g^h(\rho_{kj}^h, \hat{\theta}, \psi_k^h, k),\ p(\hat{\theta}))\}$ ▷ 3D point and its uncertainty

    end for

  end for

end for

Algorithm 2 Elevation angle MLE (Breux and Lapierre, 2020).

Input: Trained surface GP f with prior f_p; echo (ρ^h, ψ^h) of the k-th horizontal measurement; trajectory $\mathcal{H}$ and its curvilinear abscissa $\mathbf{s}$; pose $\mathbf{p}_V = [\boldsymbol{\tau}_V^T \ \mathbf{q}_V^T]^T$ and rotation axis $\mathbf{n}^v|_{\{S_V\}}$ of the vertical sonar; beam aperture b; number of samples n_q

Output: Set $\hat{\Theta}$ of local maximum likelihood elevation angles; samples $\mathcal{Q} = \{(\theta_q, \bar{\rho}_q, \sigma_q, \hat{\rho}_q)\}_{q=1}^{n_q}$ of the likelihood

$\hat{\Theta} \leftarrow \emptyset,\ \Delta_\theta \leftarrow b/n_q,\ \theta_0 \leftarrow -b/2$
for $q = 1, \dots, n_q$ **do**
  $\theta_q \leftarrow \theta_{q-1} + \Delta_\theta$
  $\mathbf{x} \leftarrow g^h(\rho^h, \theta_q, \psi^h, k)$ ▷ Candidate point, Eq. (26)
  $I \leftarrow \emptyset$
  for $i = 1, \dots, n_r - 1$ **do**
    $\mathbf{n}_i \leftarrow \mathbf{R}(\mathbf{q}_i)\mathbf{R}(\mathbf{q}_V)\mathbf{n}^v|_{\{S_V\}},\ \mathbf{n}_{i+1} \leftarrow \mathbf{R}(\mathbf{q}_{i+1})\mathbf{R}(\mathbf{q}_V)\mathbf{n}^v|_{\{S_V\}}$ ▷ Eq. (27)
    $\boldsymbol{\tau}_i^V \leftarrow \boldsymbol{\tau}_i + \mathbf{R}(\mathbf{q}_i)\boldsymbol{\tau}_V,\ \boldsymbol{\tau}_{i+1}^V \leftarrow \boldsymbol{\tau}_{i+1} + \mathbf{R}(\mathbf{q}_{i+1})\boldsymbol{\tau}_V$ ▷ Sonar head
    if $\langle \mathbf{n}_i, \mathbf{x} - \boldsymbol{\tau}_i^V \rangle \langle \mathbf{n}_{i+1}, \mathbf{x} - \boldsymbol{\tau}_{i+1}^V \rangle < 0$ **then** ▷ Eq. (28)
      $I \leftarrow I \cup \{i\}$
    end if
  end for
  $l_q^{\max} \leftarrow -\infty$ ▷ Best log-likelihood of θ_q
  for $i \in I$ **do**
    $\boldsymbol{\tau}(t) \leftarrow \boldsymbol{\tau}_i + t\,(\boldsymbol{\tau}_{i+1} - \boldsymbol{\tau}_i)$ ▷ Eq. (29)
    $[\delta\psi\ \delta\chi\ \delta\phi]^T \leftarrow \text{Euler}\left(\mathbf{R}(\mathbf{q}_i)^T\mathbf{R}(\mathbf{q}_{i+1})\right)$ ▷ Eq. (30)
    $\mathbf{R}(t) \leftarrow \mathbf{R}(\mathbf{q}_i)\,\mathbf{R}(\delta\psi\, t, \delta\chi\, t, \delta\phi\, t)$ ▷ Eqs. (31)–(32)
    $c_2, c_1, c_0 \leftarrow$ coefficients with $\mathbf{x}$, $\boldsymbol{\tau}_i$, $\boldsymbol{\tau}_{i+1}$, $\delta\boldsymbol{\omega}$ ▷ Eq. (37)
    $\hat{t} \leftarrow$ root of $c_2 t^2 + c_1 t + c_0$ in $[0, 1]$ ▷ Eqs. (34), (36)
    $\hat{\boldsymbol{\eta}} \leftarrow \left[\boldsymbol{\tau}(\hat{t})^T \ \ \mathbf{q}(\hat{t})^T\right]^T$ with $\mathbf{R}(\mathbf{q}(\hat{t})) = \mathbf{R}(\hat{t})$ ▷ Interpolated pose
    $[\hat{x}^v\ \ \hat{y}^v\ \ 0]^T \leftarrow \mathbf{x} \ominus (\hat{\boldsymbol{\eta}} \oplus \mathbf{p}_V)$ ▷ Eq. (39)
    $\hat{s} \leftarrow s_i + \|\log(\mathbf{T}_i^{-1}\hat{\mathbf{T}})\|_G$ ▷ Eq. (38)
    $\hat{\psi}^v \leftarrow \text{atan2}(\hat{y}^v, \hat{x}^v)$ ▷ Eq. (40)
    $\hat{\rho}^v \leftarrow \|[\hat{x}^v\ \ \hat{y}^v\ \ 0]^T\|_2$ ▷ Eq. (41)
    $(\bar{\rho}^*, \sigma^*) \leftarrow$ prediction of f at $(\hat{s}, \hat{\psi}^v)$, prior $f_p(\hat{\boldsymbol{\eta}}, \hat{\psi}^v)$ ▷ Eq. (42)
    $l \leftarrow -\left(\frac{(\bar{\rho}^* - \hat{\rho}^v)^2}{2\sigma^{*2}} + \ln\sigma^* + \frac{\ln 2\pi}{2}\right)$ ▷ Eq. (43)
    if $l > l_q^{\max}$ **then**
      $l_q^{\max} \leftarrow l,\ (\bar{\rho}_q, \sigma_q, \hat{\rho}_q) \leftarrow (\bar{\rho}^*, \sigma^*, \hat{\rho}^v)$ ▷ Most likely crossing
    end if
  end for
  if $q \geq 3$ **and** $l_{q-1}^{\max} > l_{q-2}^{\max}$ **and** $l_{q-1}^{\max} > l_q^{\max}$ **then** ▷ Eq. (25)
    $\hat{\Theta} \leftarrow \hat{\Theta} \cup \{\theta_{q-1}\}$
  end if
end for

Algorithm 3 Uncertainty estimate (Breux and Lapierre, 2020).

Input: Samples $\mathcal{Q} = \{(\theta_q, \bar{\rho}_q, \sigma_q, \hat{\rho}_q)\}$ of the likelihood $p(\rho^v \mid \theta)$ (Algorithm 2); local maximum $\bar{\theta} = \theta_{\bar{q}} \in \hat{\Theta}$; sampling step Δ_θ; beam aperture b; filter parameter k

Output: Distribution $p(\hat{\theta})$ of the elevation angle

$\bar{\rho}' \leftarrow (\bar{\rho}_{\bar{q}+1} - \bar{\rho}_{\bar{q}-1})/(2\Delta_\theta)$ ▷ Central finite differences
$\sigma' \leftarrow (\sigma_{\bar{q}+1} - \sigma_{\bar{q}-1})/(2\Delta_\theta)$
$\mathcal{I}(\bar{\theta}) \leftarrow \left(2\sigma'^2 + \bar{\rho}'^2\right)/\sigma_{\bar{q}}^2$ ▷ Fisher information, Eq. (50)
$\sigma_{\bar{\theta}}^2 \leftarrow 1/\mathcal{I}(\bar{\theta})$ ▷ Eq. (45)
$\sigma_l^2 \leftarrow (b + 2|\bar{\theta}|)^2(b - 2|\bar{\theta}|)\ /\ \left(4(3b + 2|\bar{\theta}|)\right)$ ▷ Unimodality, Eq. (55)
$\sigma_m \leftarrow b/(2k)$ ▷ Maximum uncertainty, Eq. (57)
$\sigma_T^2 \leftarrow \min(\sigma_l^2, \sigma_m^2)$ ▷ Eq. (58)
$\alpha \leftarrow 1,\ \beta \leftarrow 1$ ▷ Uniform distribution by default
if $\sigma_{\bar{\theta}}^2 \leq \sigma_T^2$ **then**
  $\alpha \leftarrow (b + 2\bar{\theta})\left(b^2 - 4(\bar{\theta}^2 + \sigma_{\bar{\theta}}^2)\right)/(8b\,\sigma_{\bar{\theta}}^2)$ ▷ Eq. (8)
  $\beta \leftarrow (b - 2\bar{\theta})\left(b^2 - 4(\bar{\theta}^2 + \sigma_{\bar{\theta}}^2)\right)/(8b\,\sigma_{\bar{\theta}}^2)$ ▷ Eq. (8)
end if
$p(\hat{\theta}) \leftarrow \text{Beta}(\alpha, \beta, -b/2, b/2)$ ▷ Eqs. (6), (44)

5 Results

[Present and discuss the results obtained, with numbered and titled tables and figures.]

6 Conclusions and Recommendations

6.1 Conclusions

[Summarise how the objectives stated in the introduction have been met and make some concluding remarks.]

6.2 Recommendations

[Briefly describe the proposed plans of action for the future.]

List of References

Batiot-Guilhe, P., Dörfliger, N., Ladouche, B., and Soulas, M. (2013). Contribution of a conceptual hydrogeological model to the definition of a flash flood warning system in a complex karst environment: the example of the Nîmes karst aquifer (southern France). *Engineering Geology*, 164:1–13.

Blanco, J.-L. (2010). A tutorial on SE(3) transformation parameterizations and on-manifold optimization. Technical report, University of Malaga, Malaga, Spain.

Borrmann, D., Elseberg, J., Lingemann, K., Nüchter, A., and Hertzberg, J. (2008). Globally consistent 3D mapping with scan matching. *Robotics and Autonomous Systems*, 56(2):130–142.

Breux, Y. and Lapierre, L. (2020). Elevation angle estimations of wide-beam acoustic sonar measurements for autonomous underwater karst exploration. *Sensors*, 20(14):4028.

Ford, D. C. and Williams, P. W. (2007). *Karst Hydrogeology and Geomorphology*. John Wiley & Sons, Chichester, UK.

Goldscheider, N., Chen, Z., Auler, A. S., Bakalowicz, M., Broda, S., Drew, D., Hartmann, J., Jiang, G., Moosdorf, N., Stevanovic, Z., et al. (2020). Global distribution of carbonate rocks and karst water resources. *Hydrogeology Journal*, 28(5):1661–1677.

Johnson, N. L., Kotz, S., and Balakrishnan, N. (1994). *Continuous Univariate Distributions*. John Wiley & Sons, New York, NY.

Lapierre, L., Zapata, R., Lepinay, P., and Ropars, B. (2021). Karst exploration: unconstrained attitude dynamic control for an AUV. *Ocean Engineering*, 219.

Lasbouygues, A., Louis, S., Ropars, B., Rossi, L., Jourde, H., Délas, H., Balordi, P., Bouchard, R., Dighouth, M., Dugrenot, M., Jacquemin, E., Vasseur, F., Lapierre, L., and Andreu, D. (2017). Robotic mapping of a karst aquifer. In *IFAC World Congress*.

Maréchal, B., Ladouche, B., and Dörfliger, N. (2008). Karst flash flooding in a Mediterranean environment, hydro-geological and multi-physical approach for the 3 October 1988 Nîmes flood. *Engineering Geology*, 99(3–4):168–181.

Margaritis, G. E. (2025). 3D underwater scene reconstruction with sonar data. Internship report, ENSTA – Institut Polytechnique de Paris.

Matérn, B. (1986). *Spatial Variation*, volume 36 of *Lecture Notes in Statistics*. Springer, New York, NY.

Nüchter, A., Bleier, M., Schauer, J., and Janotta, P. (2017). Improving Google's Cartographer 3D mapping by continuous-time SLAM. In *Proceedings of the 7th ISPRS International Workshop 3D-ARCH*, pages 543–549, Nafplio, Greece.

Rasmussen, C. E. and Williams, C. K. I. (2006). *Gaussian Processes for Machine Learning*. MIT Press, Cambridge, MA.